

Anycast DNS

Service technical architecture for top-level domains

Executive summary

CIRA TLD Anycast is a high-performance, high-availability secondary DNS trusted by over 490 top-level domains in the gTLD and ccTLD segments – covering nearly one-third of all top-level domains in the world. CIRA offers a 100% uptime service-level-agreement (SLA) for answering DNS queries and there have been no service-impacting outages since the service went live in December of 2014.

The service architecture combines anycast technology with multiple levels of redundancy across the entire global network. For top level domains (TLDs) we offer flexible implementation options that include using your own IP blocks and hybrid clouds that combine your local nodes with CIRA's global clouds. This technical document describes various features of the architecture and how it benefits TLDs.

"The need for secondary DNS services at the TLD level has never been clearer. CIRA is thrilled with the global response to the world-class network and service we have offered through TLD Anycast."

JACOB ZACK, DNS ARCHITECT, CIRA

A global secondary DNS service for TLDs

WHAT CIRA TLD ANYCAST OFFERS:

- The necessary aggregate bandwidth, high-capacity routers and servers, global node distribution and exchange point peering to handle authoritative resolution for a top-level domain registry.
- For domain registries the kind of reporting they need to manage a TLD business.

The CIRA universal cloud

DIVERSE INTERNET BACKBONE TRANSIT THAT DESERVES ITS OWN UNIQUE DESCRIPTOR

CIRA has structured the clouds with multiple tier-1 transit providers for optimizing performance and reliability. CIRA works constantly to grow the network of over 10,000 peers connected across dozens of peering points globally, vastly improving network performance and resiliency.

IXP connections and peering

Sites are connected to internet exchange points (IXP). In total, sites are peered with more than 10,000 networks worldwide. These peering connections are the first choice, not the last option, for networks sending traffic to CIRA – because sites are directly connected to the network, that traffic only has to cross one circuit to reach CIRA, rather than a maze of circuits around the world.

Zone propagation servers

Geographically diverse and fully redundant zone propagation server clusters in two data centers with diverse network connections accessible for IPV4 and IPV6. Only one zone propagation server needs to be operational for the service to receive zone transfers and push them out to the cloud servers.

| Global deployment

26 distinct points of presence (soon 28, with Lima and Milan coming online within the next week or so) in strategically placed and well-connected locations around the globe. Each location connects with diverse transit providers for internet connectivity. In addition, CIRA connects to multiple IXPs in each location with a focus on keeping DNS query traffic in the region. This design provides improved response time and improved resiliency for our customers. This architecture has been tested to ensure that local queries don't ping-pong across oceans when there is a better route.

Hybrid CCTLD DNS model

CIRA supports a hybrid cloud architecture that allows a TLD partner to own and operate nodes. TLD partners supply the IP address block for their cloud to effectively create a custom cloud. TLD partners can extend their local anycast infrastructure with CIRA's global nodes. This is an optional, non-standard configuration: CIRA has all the capacity and speed a TLD needs on its own, but this feature lets a partner maintain its own nodes and peering relationships domestically while using CIRA to complement its cloud. Most partners instead run a domestic cloud alongside CIRA's international cloud, or ask CIRA to deploy infrastructure locally.

Region	City	Internet Exchange Points
Africa	Lagos, Nigeria	IXPN Lagos
Africa	Cape Town, South Africa	NAP Africa Cape Town
Africa	Johannesburg, South Africa	NAP Africa Johannesburg
Asia	Delhi, India	Extreme IX Delhi
Asia	Mumbai, India	Extreme IX Mumbai
Asia	Singapore, Singapore	BBIX Singapore, Equinix Singapore, SGIX
Asia	Tokyo, Japan	BBIX Tokyo, JPIX Tokyo
Europe	Amsterdam, Netherlands	AMSIX
Europe	Frankfurt, Germany	DE-CIX Frankfurt, NL-IX
Europe	London, United Kingdom	LINX, LONAP
Europe	Madrid, Spain	Espanix Lower, Espanix Upper
Europe	Milan, Italy	MIX-IT
Europe	Paris, France	Equinix Paris, France-IX Paris
Europe	Prague, Czechia	Peering.CZ
Oceania	Melbourne, Australia	IX Australia Melbourne, MegalIX Melbourne
Oceania	Sydney, Australia	Equinix Sydney, IX Australia Sydney, MegalIX Sydney
North America	Ashburn, USA	Equinix Ashburn
North America	Calgary, Canada	YYC-IX
North America	Los Angeles, USA	Any2 Los Angeles
North America	Montreal, Canada	CANIX Montreal
North America	Queretaro, Mexico	PIT MX
North America	Toronto, Canada	TORIX
North America	Vancouver, Canada	VANIX
South America	Bogota, Colombia	PIT Colombia Bogota
South America	Lima, Peru	PIT Peru Lima
South America	Santiago, Chile	PIT Chile
South America	Sao Paulo, Brazil #1	IX.BR Sao Paulo
South America	Sao Paulo, Brazil #2	IX.BR Sao Paulo

Management transit

Each CIRA node has management transit that is separate from the internet backbone transit provider. The management transit is used to ensure a site is reachable in the event of a transit failure or a DDoS attack. Management transit is also used to update cloud server configuration and collect usage data. To increase security, a VPN is used to connect remote sites.

High availability DNS sites

CIRA's cloud is built on two node topologies. Twelve Global/Super Nodes are configured with routers that load balance DNS queries between multiple DNS servers, continuously monitor the availability of each server and remove a server from service if it becomes unavailable; together these nodes carry about 90% of all query traffic. The remaining traffic – about 10% – is handled by strategically placed IXP-only regional nodes.

Global/Super Nodes carry spare capacity that can be used in a floating way and redistributed to where it's needed.

CIRA also offers PCAP (packet capture) delivery, so that TLD customers can use their packet captures for their own internal research.

Global remote monitoring

CIRA maintains a 100+ node remote monitoring network that is constantly monitoring our availability and response times from all over the world. CIRA monitors via IPv4 and IPv6, as well as UDP and TCP, every minute, from 100+ probes around the world.

Alerting is in place to let our response team know within seconds if there's been a change to availability or round-trip times.

CIRA can produce availability and SLA reporting based on this remote monitoring network.

Management and configuration

Protected access

CIRA utilizes secure automation tools for the management of the network and TLD zones. Because of the critical nature of registry services, any changes are validated in person; CIRA does not provide customers with a self-service interface to alter their TLD configuration directly.

This is backed by a broader security program, including employee background checks, a robust HR security policy, ISO 27001-certified processes and regular third-party penetration testing. CIRA also participates in the world's leading DNS and security forums, including DNS-OARC, CENTR and ICANN.

DNS query reporting

CIRA offers read-only access to a reporting portal. CIRA can also provide PCAP files to the customer for ingestion into their own systems for those that report from multiple suppliers. At each node all DNS packet data is mirrored to a dedicated server. A Parquet/PCAP file is produced every five minutes and transported to a central site over management transit where it can be pushed to a cloud repository supplied by the customer. Data collection is done out of band so it does not impact performance.

DNS log message reporting

Log messages from the CIRA name servers that perform zone transfers into CIRA from a customer are captured, but are not currently exposed directly to customers. CIRA can provide SLA reporting on zone propagation to TLDs for monitoring.

DNS software and hardware diversity

CIRA maintains multi-platform resilience with the most recent secure versions of DNS software at all sites – including support agreements with software supplier organizations. Sites have been configured to allow rapid cut-over between applications to protect against a major zero-day event. CIRA also utilizes multiple hardware vendors to maintain hardware diversity in the same manner as we do with our DNS software. Combined, this provides protection against zero-day vulnerabilities and vendor bugs.

Protecting TLD DNS with a DDoS mitigation strategy

CIRA Anycast TLD has a 100 per cent uptime SLA for answering DNS queries. There has been no service impacting outages since the service went live in December 2014 as such, CIRA has used multiple strategies to combat the growing threat of amplified DDOS attacks.

BANDWIDTH AND QUERY CAPACITY

- Peering relationships – With over 10,000 peers globally, CIRA continues to increase the proportion of queries that can be answered over peering rather than internet transit. These IXP peering connections are the first choice, not the last option, for networks sending traffic to

CIRA. CIRA will work closely with TLDs to develop peering relationships for their regions. Traffic to CIRA has a number of incoming paths, which differentiates CIRA from some competitors who rent servers and thus have a single path in to each site.

- Packet scrubbing & filtering - CIRA has over 500 TB/sec of total packet scrubbing/filtering technology via two separate third-party DDoS mitigation providers.
- Detection and monitoring – Early detection is critical to mitigating a DDoS attack. 24/7 monitoring with automated alerts is used to detect and respond to attacks.

CIRA runs constant capacity testing and analysis, which shows that CIRA could lose whole stacks of servers and several sites at once and still be one of the fastest, most resilient, most highly-available DNS products out there. CIRA operates at under 1% of its total capacity by design, leaving room for both the expansion of customer needs and DDoS attacks.

Service component	Failure	Service impact	MTTR
Anycast name server	Name server fails	None	5 days
Anycast site	Complete data center failure	None	30 days to relocate data center
Anycast router	Complete router failure	None	5 days
Internet transit provider	Complete failure for internet transit provider that hits one cloud	Reduced capacity – queries answered by second cloud and IXPs	14 days to switch providers (worst case)
Anycast cloud	Large scale DDoS attack from offshore	Service impacting at international anycast sites	Depends on size, duration and type of attack and the efficacy of active mitigation

CIRA participates in our shared community

- DNS-OARC contributor – CIRA is an active member and contributor to DNS-OARC, including in the past and still today on relevant committees.
- Internet Engineering Task Force (IETF) – CIRA has contributed to IETF efforts to standardize DNSSEC.
- ICANN ccNSO, DNSSEC workshop, SSAC – CIRA is an active participant and contributor to the ccNSO workshops, ccNSO Tech Days, DNSSEC workshops and SSAC.

A shared commitment to growth

CIRA shares many technology, research and governance interests with our peers. By providing DNS to TLDs we aim to deepen collaboration in these areas, and to explore new product partnerships, including:

- DNS data research in machine learning research on PCAP.
- DNSSEC adoption via the registry operator - IETF draft protocol.
- Shared data center rack space to reduce costs and expand mutual global capacity.
- New product development such as internet performance testing and recursive services.

To get CIRA TLD Anycast working for your TLD, please contact us at
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